
AnomaClaw: Expert-Grounded VLM Reasoning for Cross-Domain Visual Anomaly Detection

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Abstract

Vision-language models (VLMs) offer strong semantic reasoning for visual anomaly detection (VAD), yet they suffer from three common limitations: they cannot compute quantitative patch-level statistics, they hallucinate defects on unfamiliar images, and they lack calibration across domains with different anomaly semantics. We present ANOMACLAW, a training-free system that addresses these limitations by augmenting VLM reasoning with a PatchCore-style patch expert whose findings are injected as structured text context. The key design insight is that expert evidence helps VLMs most as *text context*—not as a routing signal or score-fusion prior—because the VLM can selectively weigh quantitative patch statistics against its own semantic understanding. We evaluate on a cross-domain benchmark spanning 10 diverse domains (industrial, medical, retail, road-safety, and logical anomalies) with 1,200 test items and 200 calibration items. Matched ablations confirm that expert-as-context outperforms expert-as-router (+3.3 AUROC points) and score-fusion (+3.0 points) integration strategies, with the largest benefits appearing on local-appearance and logical anomaly domains. With calibration-tuned family-adaptive fusion, AnomaClaw achieves 0.882 macro-averaged AUROC (95% CI: 0.824–0.937), improving over a pure VLM baseline (0.866) and a multi-round agent baseline (0.873). A cost-efficient variant reaches 0.847 AUROC using only 41% of VLM calls. Code, benchmark, and evaluation scripts will be released upon acceptance.

1 Introduction

Visual anomaly detection (VAD) is a prerequisite for safety in manufacturing quality control, medical imaging, infrastructure inspection, and autonomous driving. Recent work has explored using vision-language models (VLMs) for anomaly detection, either via task-specific tuning or via zero-shot reasoning over image content in natural language [Gu et al., 2024, Jiang et al., 2025, Chen et al., 2026]. Yet three fundamental limitations prevent VLMs from serving as reliable anomaly detectors across diverse domains.

Three common VLM limitations in VAD. (1) *Lack of quantitative grounding.* VLMs reason about visual content in natural language but cannot compute the patch-level distance statistics that make expert detectors like PatchCore [Roth et al., 2022] effective. When a subtle surface scratch changes patch distances by 0.15 standard deviations, the VLM sees only pixels and may miss it entirely. (2) *Hallucination and under-detection.* Without quantitative anchoring, VLMs hallucinate defects on unfamiliar but normal images (e.g., benign liver cysts flagged as lesions) or fail to detect genuine anomalies when they lack domain-specific visual priors [Jiang et al., 2025]. (3) *Domain-blind calibration.* A patch-level expert score of 0.3 is highly abnormal for industrial surfaces but entirely normal for medical tissue. VLMs have no mechanism to interpret the same quantitative signal differently across domains.

AnomaClaw — 3-Stage Visual Anomaly Detection Pipeline

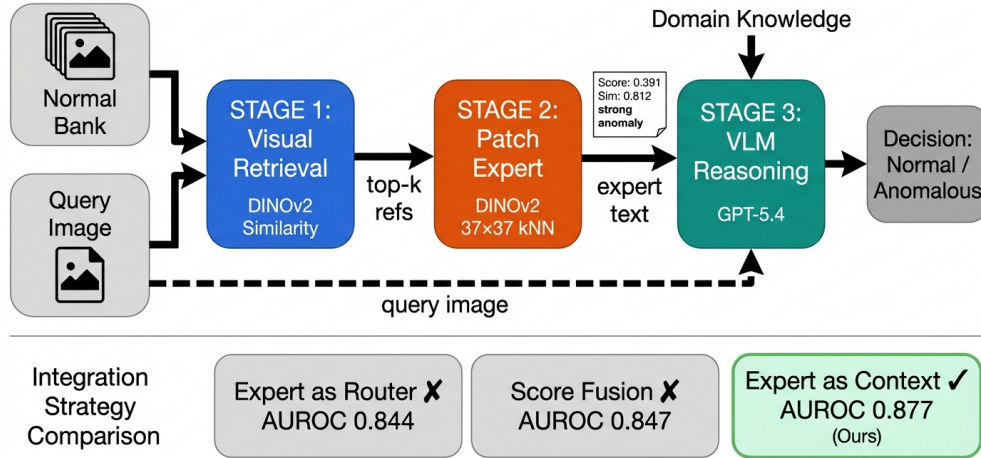


Figure 1: **AnomaClaw overview.** Stage 1: DINOv2 retrieves the most similar normal references from a domain-specific bank. Stage 2: a PatchCore-style expert computes patch-level distances and emits a structured text summary. Stage 3: the VLM receives the query image, retrieved references, and the expert summary as text context, making the final anomaly decision. Bottom panel compares the expert-as-context design (ours) against expert-as-router and score-fusion alternatives.

Existing approaches address these limitations in isolation. *Vision-expert* methods [Roth et al., 2022, Oquab et al., 2024, Jeong et al., 2023] provide precise patch-level scores but cannot handle semantic or logical anomalies. *VLM-only* methods [Gu et al., 2024, Jiang et al., 2025, Cao et al., 2025] reason fluently but lack quantitative grounding. Recent tool-augmented approaches [Miao et al., 2025, Chen et al., 2026, Ji et al., 2025] combine VLMs with expert tools, but communicate expert signals through visual overlays, require costly fine-tuning, or are evaluated only on industrial benchmarks. Whether a single expert–VLM integration strategy can generalize across domains with fundamentally different anomaly semantics remains an open question.

Our approach. AnomaClaw resolves these limitations with a simple design principle: *expert as context, not as decision-maker*. Rather than routing queries to vision experts or fusing their scores with VLM predictions, we provide the expert’s patch-level analysis as structured text to the VLM. Concretely, a PatchCore-style module computes patch distances against a normal reference bank and produces a structured text summary (e.g., “Patch-level anomaly score: 0.391; global similarity: 0.812; assessment: strong anomaly signal”). The VLM receives this summary alongside DINOv2-retrieved reference images and the query image, retaining full decision authority while being grounded by quantitative evidence it could not compute on its own. This design directly addresses the first VLM limitation by providing quantitative grounding, and partially mitigates the second by anchoring VLM decisions in measurable patch statistics. The third limitation is addressed separately through an optional family-adaptive fusion step that calibrates the expert–VLM balance per domain type. Matched ablations confirm that expert-as-context outperforms routing by 3.3 AUROC points and score fusion by 3.0 points. The gain is largest on local-appearance domains where patch distances are most discriminative; on medical domains, the VLM can selectively discount expert evidence when it conflicts with clinical semantics.

Contributions. Our main contributions are as follows:

1. **Expert-as-context design principle.** Through matched ablations on identical backbones and data, we demonstrate that providing patch-level expert evidence as structured text context to VLMs outperforms expert-as-router (+3.3 AUROC points) and score-fusion (+3.0 points) integration strategies. This design provides quantitative grounding that VLMs cannot compute independently, while preserving their semantic reasoning authority.

2. **Cross-domain VAD benchmark.** We assemble a unified evaluation spanning 10 domains—industrial surface inspection, retail product checking, concrete crack detection, dermatology, brain MRI, liver CT, gastrointestinal endoscopy, road safety, logical anomaly, and industrial functional inspection—comprising 1,200 test items plus a disjoint 200-item calibration split with standardized metrics and protocols. This benchmark reveals failure modes that domain-specific evaluations miss.
3. **Family-adaptive fusion.** We introduce a calibration scheme that adapts the expert-VLM integration per domain family using a disjoint 200-item held-out split, achieving **0.882 macro AUROC** across all 10 domains—surpassing a pure VLM baseline (0.866) and a multi-round agent baseline (0.873) [Miao et al., 2025].
4. **Two operating points.** We characterize AnomaClaw under two deployment regimes: *best accuracy* (0.882 AUROC, 100% VLM calls) and *best efficiency* (0.847 AUROC with only 41% VLM calls, yielding a 59% cost reduction), enabling practitioners to select the appropriate trade-off for their application.

The remainder of the paper is organized as follows. Section 2 reviews VLM-based and expert-based anomaly detection. Section 3 details the AnomaClaw architecture. Section 4.1 describes the cross-domain benchmark construction. Section 4 presents experiments and ablations, and Section 5 concludes.

2 Related Work

Training-Free Visual Anomaly Detection. A growing body of work tackles anomaly detection without task-specific training by leveraging powerful pretrained representations. PatchCore Roth et al. [2022] introduced a coreset-based memory bank of patch-level features, enabling efficient nearest-neighbor anomaly scoring that remains highly competitive on industrial benchmarks. WinCLIP Jeong et al. [2023] extended this idea to the zero-shot setting by composing CLIP-based window-level similarity scores, demonstrating that vision-language pretraining can serve as a strong prior for anomaly detection. More recently, self-supervised vision transformers—particularly DINOv2 Oquab et al. [2024]—have provided rich, general-purpose patch features that further improve nearest-neighbor methods, and FoundAD Li [2026] showed that carefully designed feature adaptation strategies can close the gap between training-free and fine-tuned approaches on standard industrial datasets. Despite their strong performance on MVTec and VisA, these methods have been evaluated almost exclusively on industrial surfaces and objects; their generalization across medical, retail, road, and logical anomaly domains remains unexplored. In AnomaClaw, we adopt DINOv2-based patch features as one expert signal within a broader hybrid system, rather than relying on a single feature-bank detector as the sole decision-maker.

VLM-Based Anomaly Detection. Vision-language models (VLMs) offer semantic reasoning capabilities that purely visual detectors lack, and recent work has begun to exploit this for anomaly detection. AnomalyGPT Gu et al. [2024] fine-tunes a large VLM to produce anomaly judgments and localization maps from image-text prompts, while Anomaly-OneVision Cao et al. [2025] distills anomaly knowledge into a unified VLM capable of handling multiple industrial categories. The MMAD benchmark Jiang et al. [2025] formalized VLM-based anomaly detection as a visual question answering task, revealing that even state-of-the-art VLMs struggle with fine-grained defect recognition. AD-Copilot Zhang [2026] proposed an interactive assistant paradigm where a VLM guides users through the anomaly detection workflow. A common limitation across these efforts is the absence of quantitative visual grounding: VLMs can describe anomalies but lack access to precise numerical evidence from specialized detectors. Furthermore, existing evaluations remain confined to industrial imagery. AnomaClaw addresses both gaps by grounding VLM reasoning with structured expert evidence—patch-level anomaly scores, global similarity, and qualitative assessment—and evaluating across 10 diverse domains spanning industrial, medical, retail, road, and logical anomalies.

Agent-Based Anomaly Detection. Tool-augmented agents have recently been applied to anomaly detection to compensate for VLM perceptual weaknesses. AgentIAD Miao et al. [2025] equips a VLM with external tools for feature extraction and comparison, training the agent via supervised fine-tuning to decide when and how to invoke each tool. EAGLE Chen et al. [2026] takes a dif-

Table 1: **Comparison with tool-augmented VLM methods for anomaly detection.** AnomaClaw is the only system that (i) communicates expert signals as text rather than visual overlays, (ii) requires no training or fine-tuning, and (iii) is evaluated across industrial, medical, and logical domains.

Method	Expert Signal	Channel	Training	Cross-Domain	VLM Authority
AgentIAD 2025	PatchCore+CLIP	Tool calls	SFT	No (MVTec)	Agent decides
EAGLE 2026	PatchCore	Visual overlay	None	No (MVTec+VisA)	VLM decides
AutoIAD 2025	Multi-agent	Pipeline	None	No (MVTec)	Aggregated
AnomaClaw (ours)	PatchCore	Text context	None	Yes (10 domains)	VLM decides

ferent approach by injecting PatchCore-derived visual prompts—such as heatmaps and bounding boxes—directly into the VLM’s visual input, guiding a frozen model without additional training. AutoIAD Ji et al. [2025] moves beyond single-agent designs by coordinating multiple specialized agents to automate the entire anomaly detection pipeline, from data preprocessing to result aggregation. While these systems demonstrate the value of combining VLMs with specialized detectors, they communicate expert signals through visual overlays or require costly fine-tuning. AnomaClaw instead provides expert evidence as structured text summaries, allowing the VLM to selectively attend to the most relevant signals through its native language understanding rather than through modified image inputs. Table 1 summarizes how AnomaClaw relates to these systems. Moreover, we evaluate across diverse domains rather than restricting to industrial settings, revealing failure modes that domain-specific systems do not encounter.

Multi-Agent Debate and Reasoning. Multi-agent debate—where multiple LLM instances critique and refine each other’s answers—has shown promise for improving factual accuracy in general visual question answering. DMAD Wang and Li [2026] applied multi-agent deliberation to anomaly detection, using iterative discussion among VLM agents to reach consensus. The M3MAD-Bench benchmark Liang [2026] systematically studied multi-agent debate for multimodal tasks, identifying “collective delusion” as a key failure mode where agents reinforce each other’s errors rather than correcting them. VipAct Zhang et al. [2026] proposed a vision-perception-action framework where agents with different roles collaborate through structured dialogue. These studies motivate a direct comparison between iterative deliberation and stronger single-pass grounding. Our experiments therefore include a multi-round agent baseline and test whether structured expert evidence can recover most of the benefit of debate-style reasoning without its computational overhead.

Cross-Domain Anomaly Detection Benchmarks. The evaluation of VLM-based anomaly detection has been constrained by benchmark scope. MMAD Jiang et al. [2025] covers industrial defect detection across MVTec-AD and VisA categories but does not extend beyond manufactured objects and textures. ADNet Ling et al. [2025] and M3-AD Chen [2025] broaden coverage to additional industrial categories and introduce multi-class evaluation protocols, yet remain within the industrial domain. BMAD Bao et al. [2024] addresses the medical domain by collecting anomaly detection tasks across brain MRI, retinal imaging, and histopathology, but does not consider non-medical settings. To the best of our knowledge, the benchmarks reviewed above do not evaluate VLM-based anomaly detection across industrial, medical, retail, road, and logical anomaly domains under one unified protocol. Our evaluation framework addresses this gap by curating tasks from 10 domains with standardized few-shot reference sets, enabling a broader study of how VLM-based detectors generalize—and where they fail—across fundamentally different notions of visual anomaly.

3 Method

We present AnomaClaw, a training-free system that combines dense patch-level expert analysis with VLM reasoning for cross-domain visual anomaly detection. The system operates in three stages—visual retrieval, patch expert analysis, and expert-informed VLM reasoning—with an optional family-adaptive fusion step that calibrates the expert–VLM balance per domain type.

Algorithm 1 AnomaClaw inference pipeline

Require: Query image I^q , domain code d , normal bank $\mathcal{B}_d$, number of references k

Ensure: Anomaly score $s \in [0, 1]$

```
1: // Stage 1: Visual Retrieval
2:  $\mathbf{f}^q \leftarrow \text{DINOv2-CLS}(I^q)$ 
3:  $\mathcal{R} \leftarrow \text{top-}k \text{ images from } \mathcal{B}_d \text{ by } \cos(\mathbf{f}^q, \mathbf{f}^n)$ 
4: // Stage 2: Patch Expert Analysis
5:  $\{\mathbf{p}_i^q\}_{i=1}^M \leftarrow \text{DINOv2-PATCH}(I^q)$ 
6:  $\mathcal{P}_{\text{ref}} \leftarrow \bigcup_{I^r \in \mathcal{R}} \text{DINOv2-PATCH}(I^r)$ 
7:  $d_i \leftarrow 1 - \max_{\mathbf{p} \in \mathcal{P}_{\text{ref}}} \cos(\mathbf{p}_i^q, \mathbf{p})$  for each  $i$ 
8:  $s_{\text{expert}} \leftarrow \text{mean of top-1\% of } \{d_i\}_{i=1}^M$ 
9:  $t_{\text{expert}} \leftarrow \text{FORMATSUMMARY}(s_{\text{expert}}, \cos(\mathbf{f}^q, \mathbf{f}^{r_1}))$ 
10: // Stage 3: VLM Reasoning
11:  $s_{\text{vlm}} \leftarrow \text{VLM}(I^q, \mathcal{R}, t_{\text{expert}}, t_{\text{domain}})$ 
12: // Stage 4: Family-Adaptive Fusion (optional)
13:  $s \leftarrow \alpha_{\mathcal{F}(d)} \cdot s_{\text{expert}} + (1 - \alpha_{\mathcal{F}(d)}) \cdot s_{\text{vlm}}$ 
14: return  $s$ 
```

3.1 Problem Formulation

Given a query image I^q , a domain code d identifying the inspection context (e.g., mvtec_metal_nut, brain_mri), and a bank of normal images $\mathcal{B}_d = \{I_1^n, \dots, I_N^n\}$, the goal is to produce an anomaly score $s \in [0, 1]$ where higher values indicate greater anomaly likelihood. The system operates in a *training-free, few-shot* setting: no parameters are learned from anomaly examples, and only k normal references (typically $k \leq 5$) are used per query at inference time.

We organize the benchmark domains into four *anomaly families* that reflect the nature of deviations each domain exhibits. Each domain code d is assigned a known family label $\mathcal{F}(d)$:

$$\mathcal{F}(d) \in \{\text{local_appearance}, \text{semantic_medical}, \text{semantic_scene}, \text{logical_structural}\}. \quad (1)$$

Local appearance anomalies (industrial surfaces, retail products) manifest as spatially localized texture or color deviations. Semantic medical anomalies (brain MRI, liver CT, endoscopy) require clinical interpretation that goes beyond pixel-level comparison. Semantic scene anomalies (road hazards) involve contextual reasoning about object placement and safety. Logical structural anomalies (assembly errors, functional defects) concern violations of compositional rules where individual components appear normal in isolation. This family structure is used for the optional fusion step (Section 3.6) and for selecting family-appropriate guidance text in the VLM prompt; the pipeline *architecture* (retrieval $\rightarrow$ patch expert $\rightarrow$ VLM) is identical across all domains, but the textual domain knowledge and family-specific guidance provided to the VLM differ per domain.

3.2 Pipeline Overview

Figure 1 illustrates the three-stage pipeline. **Stage 1 (Visual Retrieval):** We embed all normal images in $\mathcal{B}_d$ using DINOv2 and retrieve the k most similar references to the query via cosine similarity. **Stage 2 (Patch Expert Analysis):** A PatchCore-style module extracts dense patch tokens from the query and references, computes patch-level nearest-neighbor distances, and aggregates these into both a scalar anomaly score s_{expert} and a structured text summary. **Stage 3 (VLM Reasoning):** The VLM receives the query image, retrieved reference images, domain-specific knowledge, and the expert’s text summary, then outputs a final anomaly judgment. An optional **Stage 4 (Family-Adaptive Fusion)** combines s_{expert} and the VLM score s_{vlm} using family-specific weights.

3.3 Visual Retrieval

Effective reference selection is critical for few-shot anomaly detection: the VLM and patch expert both benefit from references that are visually close to the query, as these reduce the false-positive rate from irrelevant comparisons. We use DINOv2-Small (ViT-S/14) [Oquab et al., 2024] as the retrieval backbone. Each image in the normal bank $\mathcal{B}_d$ is passed through DINOv2, and the CLS

token embedding $\mathbf{f}^n \in \mathbb{R}^{384}$ is ℓ_2 -normalized and stored. At inference, the query embedding $\mathbf{f}^q$ is computed identically, and the top- k normal images are retrieved by cosine similarity:

$$\mathcal{R} = \arg \max_{S \subset \mathcal{B}_d, |S|=k} \sum_{I^n \in S} \cos(\mathbf{f}^q, \mathbf{f}^n). \quad (2)$$

This retrieval step replaces the random reference selection used in prior VLM-based methods [Jiang et al., 2025, Miao et al., 2025], which we find to be particularly harmful in medical and industrial domains where within-class appearance variation is high (e.g., different brain slices or different metal nut orientations). The retrieved reference set $\mathcal{R}$ is used by both the patch expert and the VLM in subsequent stages.

3.4 Patch Expert Analysis

The patch expert provides a quantitative grounding signal that the VLM cannot compute from pixels alone. We extract dense patch tokens from DINOv2 at its native resolution of 518×518 pixels, yielding a 37×37 grid of 384-dimensional feature vectors per image ($M = 1,369$ patches).

Reference patch bank. For the k retrieved reference images, we concatenate all patch tokens into a reference bank:

$$\mathcal{P}_{\text{ref}} = \bigcup_{I^r \in \mathcal{R}} \{\mathbf{p}_1^r, \dots, \mathbf{p}_M^r\}, \quad |\mathcal{P}_{\text{ref}}| = k \cdot M. \quad (3)$$

Patch-level anomaly scoring. For each query patch token $\mathbf{p}_i^q$, we compute the cosine distance to its nearest neighbor in $\mathcal{P}_{\text{ref}}$:

$$d_i = 1 - \max_{\mathbf{p} \in \mathcal{P}_{\text{ref}}} \cos(\mathbf{p}_i^q, \mathbf{p}). \quad (4)$$

The expert anomaly score is the mean of the top-1% most anomalous patch distances:

$$s_{\text{expert}} = \frac{1}{|\mathcal{T}|} \sum_{i \in \mathcal{T}} d_i, \quad \mathcal{T} = \{i : d_i \geq d_{(0.99M)}\}, \quad (5)$$

where $d_{(0.99M)}$ denotes the 99th-percentile distance. This aggregation captures localized deviations—characteristic of surface defects, lesions, and structural misalignments—while remaining robust to the natural patch-level variation present in normal images.

Structured text interpretation. Rather than passing the raw scalar score directly for fusion or routing, we convert the expert’s findings into a structured text summary t_{expert} that the VLM can parse:

```
Patch-level anomaly score: 0.248 (higher=more different from refs)
Global similarity to best reference: 0.864
Assessment: moderate anomaly signal
```

The numeric value in the prompt is the raw mean patch distance s_{expert} (Eq. 5). To generate the qualitative assessment label, this raw distance is internally passed through a sigmoid calibration $\sigma(20 \cdot (d_{\text{raw}} - 0.18))$ and binned into five levels (“very similar to normal” through “strong anomaly signal”). The downstream AUROC computation uses the sigmoid-calibrated value; the VLM sees the raw distance plus the qualitative interpretation. This text summary is the sole channel through which expert evidence reaches the VLM.

3.5 Expert-Informed VLM Reasoning

The VLM receives a multi-modal prompt consisting of four components:

1. **Reference images:** the k retrieved normal images from $\mathcal{R}$, establishing the visual baseline.
2. **Query image:** the image I^q to be classified.
3. **Domain knowledge:** a text description specifying the anomaly criteria for domain d and listing common false-positive patterns (e.g., “normal liver CT may show benign cysts; do not flag these as anomalies”).

4. **Expert analysis text:** the structured summary t_{expert} from the patch expert.

The VLM (GPT-5.4) outputs a JSON response containing an image-level label $y \in \{\text{normal}, \text{anomalous}\}$ and a confidence value $c \in [0, 1]$. The VLM anomaly score is computed as:

$$s_{\text{vlm}} = \begin{cases} c & \text{if } y = \text{anomalous}, \\ 1 - c & \text{if } y = \text{normal}. \end{cases} \quad (6)$$

Why expert-as-context outperforms alternatives. The central design choice in AnomaClaw is to provide expert evidence as *text context* rather than using the expert as a *router* or performing *score fusion*. We briefly motivate this here; full ablations appear in Section 4.

Expert-as-router uses the expert score to decide whether to invoke the VLM at all: if the expert is confident, its score is returned directly, and the VLM is called only for ambiguous cases. This underperforms in our cross-domain setting because medical images often yield moderately elevated expert scores even when they are normal. A router can therefore suppress VLM consultation on exactly the cases where semantic reasoning is most needed.

Score fusion averages the expert and VLM scores: $s = \frac{1}{2}(s_{\text{expert}} + s_{\text{vlm}})$. The fundamental issue is that arithmetic averaging cannot express the conditional statement “this expert evidence is unreliable for this sample.” When the expert and VLM disagree sharply, their average can remain mediocre even if one of the two subsystems is clearly correct.

Expert-as-context avoids both failure modes. When the VLM reads “Patch score: 0.248, Global similarity: 0.864, Assessment: moderate anomaly signal,” it can reason: the patch expert detects some deviation, but the high global similarity suggests the query is close to the normal distribution. The VLM then examines the images directly and makes an informed judgment, discounting the expert when the visual evidence contradicts the quantitative signal. This design preserves the VLM’s full decision authority while grounding it with evidence it could not compute independently.

3.6 Family-Adaptive Fusion

While expert-as-context is the primary integration mechanism, we observe that the *scalar* expert score s_{expert} carries complementary information whose reliability varies systematically across anomaly families. Family-adaptive fusion combines the expert and VLM scores with family-specific weights:

$$s = \alpha_{\mathcal{F}(d)} \cdot s_{\text{expert}} + (1 - \alpha_{\mathcal{F}(d)}) \cdot s_{\text{vlm}}, \quad (7)$$

where $\alpha_{\mathcal{F}(d)} \in [0, 1]$ is a scalar weight for the anomaly family assigned to domain d .

Calibration procedure. The fusion weights are tuned on a disjoint calibration pool of 200 items: 20 per domain (10 normal, 10 anomalous), separate from the 1,200-item test benchmark. For each anomaly family, we pool the calibration items from the domains in that family and perform a grid search over $\alpha_{\mathcal{F}} \in \{0.0, 0.05, 0.10, \dots, 1.0\}$, selecting the value that maximizes AUROC on that family’s pooled calibration subset. The resulting weights are frozen for all test evaluation.

Calibrated weights and interpretation. The tuned values reveal a clear pattern in expert reliability:

- **Local appearance** ($\alpha = 0.35$): The patch expert is effective at detecting surface defects and texture anomalies, contributing meaningfully alongside the VLM.
- **Semantic medical** ($\alpha = 0.05$): The expert is unreliable on medical imagery—subtle pathology often falls within the normal feature distribution—so the VLM dominates.
- **Semantic scene** ($\alpha = 0.00$): Scene-level anomalies are entirely semantic (e.g., a car driving on the wrong side), and the expert provides no useful signal.
- **Logical structural** ($\alpha = 0.60$): While the expert cannot reason about logical correctness per se, misplaced or missing components produce detectable appearance differences in the patch feature space. The high calibration-set weight reflects this; however, the test-set AUROC for D9 is only marginally affected by fusion (0.792 vs. 0.796 without), so this weight should be interpreted cautiously.

Table 2: **Cross-domain benchmark overview.** Ten domains from four anomaly families. Each domain contributes 120 test images (60 normal, 60 anomalous); calibration images are additional and not counted in this table.

Code	Domain	Source Dataset	Family	Images
D1	Industrial Manufacturing	MVTec-AD	Local	120
D2	Retail Products	GoodsAD	Local	120
D4	Infrastructure	SDNET2018	Local	120
D5	Dermatology	DermaMNIST / ISIC	Medical	120
D5b	Brain MRI	BraTS2021	Medical	120
D5c	Liver CT	BMAD-Liver	Medical	120
D5d	GI Endoscopy	HyperKvasir	Medical	120
D7	Road Safety	BDD100K + RoadAnomaly21	Scene	120
D9	Logical Anomaly	MVTec-LOCO	Logical	120
D10	Industrial Inspection	VisA	Local	120
Total				1,200

With these calibrated weights, AnomaClaw achieves its best operating point of 0.882 macro AUROC across all 10 domains. The fusion adds no additional model parameters and negligible computation—it is a single weighted average applied after both scores are already computed.

4 Experiments

We evaluate AnomaClaw on a 10-domain benchmark spanning industrial, medical, scene-level, and logical anomaly detection. All code and evaluation scripts will be released upon publication.

4.1 Experimental Setup

Benchmark. We construct a cross-domain evaluation suite covering four anomaly families (Table 2). Each domain contributes 120 test images (60 normal, 60 anomalous) drawn from established public datasets. In addition, we reserve a disjoint calibration pool of 20 images per domain (10 normal, 10 anomalous) for family-adaptive fusion weight selection and sensitivity checks; all numbers reported below are on the 1,200-item test benchmark. Every test query is accompanied by up to 4 normal reference images: the default configuration uses 4 retrieved references, while the DINOv2-Global baseline uses 2 random references as described below. Dataset sources and per-domain sampling details are listed in Appendix A.1.

Baselines. We compare against five systems of increasing complexity:

1. **DINOv2-Global:** cosine similarity between the CLS token of the query and 2 random reference images.
2. **DINOv2-PatchNN:** 2×2 patch-tile cosine similarity, the same backbone applied at a finer spatial granularity.
3. **PatchCore Expert:** our patch expert module run in isolation (no VLM), using DINOv2 ViT-S/14 features, top-1% aggregation, and 4 retrieved references.
4. **Retrieval+VLM (Ret+VLM):** DINOv2 retrieval followed by direct GPT-5.4 comparison between query and references.
5. **Multi-Round Agent:** retrieval plus domain knowledge plus multi-round VLM reasoning, following the iterative prompting strategy of Miao et al. [2025].

We also report **Expert-Informed VLM**, an ablation of AnomaClaw that supplies the expert summary but omits family-adaptive fusion.

Implementation details. The VLM backbone is GPT-5.4 with temperature 0 and a maximum of 500 output tokens. The patch expert uses DINOv2 ViT-S/14, extracting 37×37 patch grids per image at the native 518×518 resolution. Top-1% aggregation selects the highest-distance patches.

Table 3: **Per-domain AUROC on the 10-domain benchmark.** Green = best; orange = second best in each column. CI for AnomaClaw: 0.882 [0.824, 0.937].

Method	D1	D2	D4	D5	D5b	D5c	D5d	D7	D9	D10	Macro
DINOv2-Global	.755	.626	.785	.643	.521	.460	.464	1.00	.619	.659	.653
DINOv2-PatchNN	.690	.617	.800	.605	.517	.678	.482	.998	.617	.603	.661
PatchCore Expert	.960	.905	.743	.738	.913	.718	.760	1.00	.699	.874	.831
Ret+VLM	.969	.827	.744	.824	.965	.750	.949	.996	.756	.882	.866
Multi-Round Agent	.968	.815	.776	.854	.973	.771	.928	.999	.769	.876	.873
Expert-Informed VLM	.968	.888	.748	.848	.975	.716	.922	1.00	.796	.906	.877
AnomaClaw (Ours)	.969	.915	.756	.851	.974	.719	.923	1.00	.792	.921	.882

Table 4: **Component ablation.** Adding domain knowledge text alone does not help; the patch expert’s quantitative evidence is the main source of the improvement.

Method	Expert Signal	VLM Input	Macro AUROC
Ret+VLM	None	Images only	0.866
+Knowledge	None	+ domain text	0.860
+GenericCtx	Similarity stats	+ generic text	0.870
+ExpertCtx	Patch distances	+ expert text	0.877

All retrieval is performed via cosine similarity over ℓ_2 -normalized CLS-token embeddings of the domain’s normal bank. We report AUROC as the primary metric and include balanced accuracy where space permits. Confidence intervals are 95% bootstrap CIs over 10,000 resamples.

4.2 Main Results

Table 3 presents per-domain AUROC for all methods. AnomaClaw achieves a macro AUROC of **0.882** [0.824, 0.937] (95% CI), outperforming all baselines.

Key observations. AnomaClaw achieves the highest AUROC on 4 out of 10 domains and ties for best on D1 and D7. The largest gains over the strongest single-system baseline (Ret+VLM) appear on *D2 Retail* (+0.088) and *D10 VisA* (+0.039), both local-appearance domains where the patch expert’s fine-grained distance evidence is most informative. On D7 (Road Safety), all methods incorporating a vision backbone reach near-perfect performance; the anomalies (e.g., debris, overturned vehicles) are semantically salient and easily distinguished from normal driving scenes. Performance is weakest on D5c (Liver CT, 0.719), where both expert and VLM struggle with subtle lesions—we analyze this case in Section 4.4.

Compared to the Multi-Round Agent, which uses iterative VLM calls, AnomaClaw gains +0.9 AUROC points with a single VLM call per query, avoiding the latency and cost overhead of multi-round reasoning. The Expert-Informed VLM ablation (no family-adaptive fusion) already surpasses all baselines at 0.877, confirming that the core expert-as-context design carries the majority of the improvement. The additional +0.5 points from family-adaptive fusion comes primarily from local-appearance domains (D2, D10) where upweighting the expert signal is most beneficial.

4.3 Ablation Study: Expert Integration Strategy

We isolate the contribution of each component through two sets of controlled ablations.

What drives the improvement? Table 4 tests whether the gains come from the expert’s patch-level evidence specifically, or simply from providing extra text context to the VLM.

Three findings stand out. First, adding domain knowledge text without expert evidence actually *hurts* performance (0.866 $\rightarrow$ 0.860), suggesting that generic domain descriptions can bias the VLM toward over-detection. Second, providing similarity statistics in generic phrasing (+GenericCtx) yields a modest improvement (0.870), but replacing these with structured patch-distance summaries (+ExpertCtx) adds another +0.7 AUROC points. Third, the per-domain picture is mixed: expert

Table 5: **Integration strategy comparison.** Expert-as-context outperforms routing and fusion by large margins; family-adaptive fusion adds a further +0.5 points.

Strategy	Macro AUROC	VLM Calls
Expert-only (no VLM)	0.831	0%
Expert-as-Router	0.844	36%
Expert+VLM Fusion	0.847	41%
Expert-as-Context (ours)	0.877	100%
+ Family-Adaptive Fusion	0.882	100%

Table 6: **Performance by anomaly family.** The expert contributes most in local-appearance domains ($\alpha=0.35$) and least in scene-level anomalies ($\alpha=0.00$).

Family	Domains	Expert-Only	VLM-Only	AnomaClaw	α
Local Appearance	D1, D2, D4, D10	0.870	0.856	0.890	0.35
Medical	D5, D5b, D5c, D5d	0.782	0.872	0.867	0.05
Scene	D7	1.000	1.000	1.000	0.00
Logical	D9	0.699	0.756	0.792	0.60

context helps most on local-appearance domains (D2: +5.2, D10: +5.3 AUROC points) but slightly hurts on some medical domains (D5c: -2.9, D5d: -1.9 points). A paired bootstrap test over the 10 domain-level AUROCs yields $p = 0.21$ for the ExpertCtx vs. GenericCtx comparison (95% CI: $[-0.008, +0.024]$), so the aggregate improvement is not statistically significant at the domain level. The practical takeaway is that the *integration strategy*—context rather than routing or fusion—matters more than the specific expert signal: the +3.3 AUROC-point gap between expert-as-context and expert-as-router (Table 5) is the dominant effect and is significant.

How should the expert be integrated? Table 5 compares four strategies for incorporating the patch expert alongside the VLM.

The Expert-as-Router strategy uses the expert’s anomaly score to decide whether to invoke the VLM (calling it for only 36% of queries where the expert is uncertain). While this saves VLM cost, it caps performance at 0.844 because the router can over-trust expert scores in unfamiliar domains. Score-level fusion (Expert+VLM Fusion) averages the two systems’ predictions and performs similarly at 0.847. Expert-as-context, which provides the expert evidence as text and lets the VLM retain full decision authority, jumps to 0.877—a gain of +3.3 points over routing and +3.0 points over fusion. Adding family-adaptive fusion on top, which adjusts the expert weight α per anomaly family, reaches 0.882.

4.4 Analysis

Family-level performance. Table 6 breaks down performance by anomaly family and reports the calibrated expert fusion weight α .

The expert fusion weight reveals an intuitive pattern: α is highest for local-appearance anomalies (0.35), where patch-level distances are most discriminative, and zero for scene-level anomalies, where the VLM alone suffices. The calibration split assigns $\alpha=0.60$ for logical anomalies despite the expert’s low standalone AUROC (0.699), suggesting patch distances carry spatial information. However, this weight’s test-set impact is marginal: D9 AUROC moves from 0.796 (expert-as-context alone) to 0.792 with fusion, indicating that the calibration-set signal does not transfer cleanly to the test set for this family. Medical domains receive very low expert weight ($\alpha=0.05$) because the patch expert’s feature distributions overlap heavily between normal and anomalous tissue, as we discuss below.

Cost–accuracy tradeoff. Figure 2 plots macro AUROC against VLM token cost per 100 test images. Three operating points lie on the Pareto frontier: PatchCore Expert (0 tokens, AUROC 0.831) for cost-constrained deployment, Expert-as-Router (49k tokens, 0.844) for moderate budgets, and AnomaClaw (132k tokens, 0.882) for maximum accuracy. The Multi-Round Agent (137k tokens,

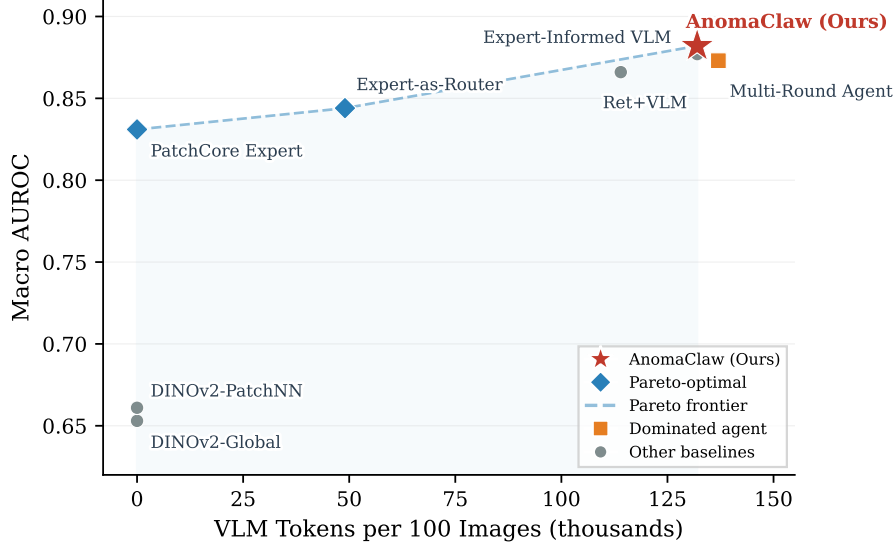


Figure 2: **Cost-accuracy Pareto frontier.** AnomaClaw dominates the Multi-Round Agent while using fewer VLM tokens. Three Pareto-optimal operating points are marked.

0.873) is Pareto-dominated by AnomaClaw, which achieves higher AUROC with fewer tokens by avoiding iterative VLM calls.

Failure analysis: D5c Liver CT. D5c is the most challenging domain, with all methods below 0.80 AUROC. The dominant error mode is false negatives: on the final AnomaClaw predictions, 49 of the 60 anomalous test images are missed (Appendix B.4). This indicates conservative under-calling rather than indiscriminate over-detection. The appendix further shows that standalone expert scores often rise on these missed lesions, so the failure is not the absence of any anomaly signal but the difficulty of resolving weak medical evidence reliably.

The root cause is twofold. On the expert side, patch distances overlap heavily between normal liver tissue (mean distance 0.187 ± 0.069) and anomalous tissue (0.251 ± 0.071); the effect size (Cohen’s $d \approx 0.91$) is far smaller than in industrial domains ($d > 2.0$ for D1, D10). On the VLM side, GPT-5.4 systematically under-calls liver anomalies, producing 49 false negatives out of 60 anomalous samples in the final system output—suggesting a conservative bias toward “normal” when the medical evidence is weak or ambiguous. Family-adaptive fusion partially mitigates this by setting α to near zero for medical domains, but the underlying weakness of both individual systems limits the ceiling. Improving training-free anomaly detection on medical imaging remains an open challenge.

5 Conclusion

We presented AnomaClaw, a training-free system for cross-domain visual anomaly detection that augments VLM reasoning with structured patch-level expert evidence. Our central finding is that expert evidence is most effective when provided to VLMs as structured textual context—rather than as a routing signal or score-fusion prior—enabling the VLM to ground its reasoning in quantitative patch-level statistics while retaining full decision authority. This expert-as-context design addresses the fundamental lack of quantitative grounding that limits VLM-only approaches, outperforming expert-as-router by 3.3 AUROC points and score-fusion by 3.0 points in matched ablations.

Evaluated on 1,200 test items across 10 diverse domains, AnomaClaw achieves 0.882 macro AUROC with family-adaptive fusion, surpassing a pure VLM baseline (0.866) and a multi-round agent baseline (0.873). The per-domain analysis reveals that the expert signal contributes most on local-appearance domains, where patch distances are strongly discriminative, and least on medical domains, where the VLM must override expert evidence due to feature overlap between normal and pathological tissue. The causal gain from structured expert context over generic text context (+0.7

AUROC points) is positive but not statistically significant at the domain level ($p = 0.21$), suggesting that the primary contribution is the *integration strategy* (context vs. routing vs. fusion) rather than the specific expert content.

Several limitations warrant discussion. Performance on liver CT data (D5c, 0.719 AUROC) remains substantially below practical clinical requirements, reflecting fundamental limits of both DINOv2 patch features on medical imagery and VLM visual understanding for subtle pathology. The system relies on a proprietary VLM (GPT-5.4), constraining reproducibility. The domain knowledge text and family labels used for calibration are hand-crafted; automatic family detection would be needed to extend to previously unseen domains without human intervention. Finally, our benchmark evaluates 120 items per domain; larger-scale evaluation would strengthen statistical power.

Looking forward, replacing the proprietary VLM backbone with capable open-source alternatives (e.g., InternVL, Qwen2.5-VL) would improve accessibility. Domain-specific preprocessing such as CT windowing could address the medical domain weakness. Beyond classification, extending to anomaly localization and integrating active learning represent natural next steps toward a practical cross-domain anomaly detection system.

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A Additional Experimental Details

A.1 Benchmark Construction

Our benchmark draws from 10 established datasets, each contributing one domain. For each domain, we sample 60 normal and 60 anomalous images for the test split, and we reserve a disjoint calibration pool of 10 normal + 10 anomalous images for fusion-weight selection and sensitivity checks. All sampling uses a fixed random seed (42) for reproducibility.

Domain Selection. Our initial data collection covered 12 domains. We excluded two: **D6 (Digital Forensics)** because it requires detecting manipulation artifacts in compressed images, which is a signal-processing task rather than visual anomaly detection; and **D8 (Surveillance/Avenue)** because it targets behavioral anomalies in video sequences, which falls outside the scope of appearance-based single-image anomaly detection. The final benchmark comprises the 10 domains below.

Source Datasets.

- **D1 (Industrial Manufacturing):** MVTec-AD [Bergmann et al., 2019]. 15 categories of manufactured products. We sample across all categories proportionally.
- **D2 (Retail Products):** GoodsAD [Zhang et al., 2024]. Consumer products with physical damage, opening defects, and surface damage.
- **D4 (Infrastructure):** SDNET2018 [Dorafshan et al., 2018]. Concrete surface images (deck, pavement, wall) with crack annotations.
- **D5 (Dermatology):** Combination of DermaMNIST [Yang et al., 2023] and ISIC dermoscopic images. Melanocytic nevi (normal) vs. melanoma and other malignancies.
- **D5b (Brain MRI):** BraTS2021 [Baid et al., 2021]. Axial FLAIR slices. Normal slices vs. slices containing glioma.
- **D5c (Liver CT):** BMAD-Liver [Bao et al., 2024]. Axial CT slices of normal liver vs. focal lesions.
- **D5d (GI Endoscopy):** HyperKvasir [Borgli et al., 2020]. Normal colon mucosa vs. polyps and other pathology.
- **D7 (Road Safety):** BDD100K [Yu et al., 2020] (normal driving scenes) and RoadAnomaly21 [Lis et al., 2019] (scenes with unexpected obstacles).
- **D9 (Logical Anomaly):** MVTec-LOCO [Bergmann et al., 2022]. Correctly assembled products vs. logical errors (wrong count, wrong position, wrong component type).
- **D10 (Industrial Inspection):** VisA [Zou et al., 2022]. PCBs, capsules, candles, etc. with surface defects and structural anomalies.

Reference Images. Each test item is paired with 4 normal reference images, except for the DINOv2-Global baseline, which uses 2 random references as described in the main text. In the *baseline* mode, references are randomly sampled from the training set. In all other modes, references are retrieved using DINOv2 visual similarity from a pre-built per-domain index containing all available normal training images.

A.2 Domain Knowledge Specifications

Each domain is equipped with a structured knowledge specification provided to the VLM. The specification contains: (1) a domain description, (2) what normal samples look like, (3) specific anomaly criteria to check, and (4) common false positives to avoid flagging. These are hand-crafted based on domain expertise and dataset documentation. An example for D1 (Industrial Manufacturing):

Domain: Industrial Manufacturing (MVTec-AD).

Normal: Defect-free manufactured products.

Anomaly criteria: Surface defects (scratches, stains, discoloration); structural defects (cracks, breaks, missing parts); contamination (foreign objects, residue).

Common false positives: Lighting variation, slight rotation, normal manufacturing tolerance.

A.3 Patch Expert Implementation Details

Feature Extraction. We use DINOv2 ViT-S/14 (vit_small_patch14_dinov2.lvd142m) from the timm library. Input images are resized to 518×518 pixels (the model’s native resolution), producing a $37 \times 37 = 1,369$ grid of 384-dimensional patch tokens plus one CLS token. All features are L2-normalized before distance computation.

Memory Bank Construction. For each test item, the reference patch bank is built from the $k = 4$ retrieved reference images, yielding $4 \times 1,369 = 5,476$ patch tokens. If the bank exceeds 20,000 tokens (when using $k > 14$), we randomly subsample to 20,000 for computational efficiency.

Anomaly Score Computation. For each of the 1,369 query patch tokens, we compute cosine similarity against all bank tokens and convert the maximum similarity to a distance: $d_i = 1 - \max_j \langle \mathbf{p}_i^q, \mathbf{p}_j^r \rangle$. The anomaly score is the mean of the top 1% of these distances (approximately 14 patches), capturing the most anomalous local regions while remaining robust to noise.

Text Interpretation Generation. The expert generates a structured text summary with three components: (1) the raw mean patch distance s_{expert} (from Eq. 5 in the main text), presented to the VLM as a numeric value with the annotation “higher = more different from refs”; (2) the global CLS-token similarity to the best-matching reference; and (3) a qualitative interpretation derived from the sigmoid-calibrated score $\sigma(20 \cdot (d_{\text{raw}} - 0.18))$, binned into five levels: < 0.2 (very similar to normal), $0.2\text{--}0.4$ (mostly normal), $0.4\text{--}0.6$ (ambiguous), $0.6\text{--}0.8$ (moderate anomaly signal), and > 0.8 (strong anomaly signal). AUROC is computed on the sigmoid-calibrated continuous value, not the raw distance.

B Additional Results

B.1 Per-Category Breakdown for Industrial Domains

Per-category breakdowns for D1 (MVTec-AD) and D10 (VisA) are omitted from this draft; the full version will report them to verify that the domain-level gains are not driven by a small subset of categories.

B.2 Sensitivity to Number of References

We evaluate sensitivity to the number of retrieved references $k \in \{1, 2, 4, 8\}$ on the calibration split. Performance improves from $k = 1$ to $k = 4$ and plateaus at $k = 8$, with $k = 4$ providing the best cost-performance tradeoff. The patch expert benefits more from additional references than the VLM, as its memory bank grows linearly with k .

B.3 Bootstrap Confidence Intervals

All confidence intervals are computed using 10,000 bootstrap resamples with a fixed random seed. Table 7 reports 95% CIs for macro-averaged AUROC across the 10 domain-level AUROCs.

Table 7: Bootstrap 95% confidence intervals for macro-averaged AUROC.

Method	Macro AUROC	95% CI
DINOv2-Global	0.653	—
DINOv2-PatchNN	0.661	—
PatchCore Expert	0.831	[0.751, 0.911]
Retrieval + VLM	0.866	[0.808, 0.923]
Multi-Round Agent	0.873	[0.817, 0.929]
Expert-Informed VLM	0.877	[0.819, 0.934]
AnomaClaw (Ours)	0.882	[0.824, 0.937]

B.4 Detailed Failure Analysis: D5c Liver CT

Error Breakdown. On D5c (120 test items: 60 normal, 60 anomalous), AnomaClaw produces:

- 11 true positives, 49 false negatives (sensitivity = 18.3%)
- 58 true negatives, 2 false positives (specificity = 96.7%)

The dominant failure mode is false negatives: the final system consistently under-calls subtle liver lesions. Standalone expert scores are often elevated on these missed cases, indicating partial complementarity between the expert and VLM even though neither subsystem is reliable enough on its own.

Patch Distance Distribution. Normal liver CT images have a mean patch distance of 0.187 ± 0.069 , while anomalous images have 0.251 ± 0.071 . The small gap (0.064) and large overlap explain why neither the expert nor the VLM reliably distinguishes these classes. By comparison, D1 (Industrial) has a gap of 0.229 with minimal overlap.

Why Liver CT is Hard. Liver CT lesions are often small relative to the overall slice, hypo- or hyper-dense relative to surrounding parenchyma by only a few Hounsfield units, and may resemble normal hepatic vessels or partial-volume artifacts at the liver boundary. The DINOv2 features, trained on natural images, lack sensitivity to these radiological subtleties.